

A Note on the Arrhenius Plot

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You have been told that when using the Arrhenius equation the log A value is less than 13.2 when the entropy of activation is negative and greater than 13.2 when it is positive.¹ What is the justification for that rule-of-thumb? This discussion will explore that idea.

Interpreting the Pre-exponential A -factor

The Arrhenius equation² relates the temperature to the rate of a reaction using the empirical values A and E_a . The latter can be considered a measure of “activation energy” and the former is a fudge-factor that accounts for all manner of other aspects of the reaction, including the entropy of activation. This equation assumes no knowledge of the process (it was created before chemists knew about transition states) and is often used to predict reaction rates at various temperatures. But, it only gives hints about the enthalpy and entropy changes in the process.

The Eyring equation³ is used for single steps in a reaction. If you can determine the rate constants for a single-step in a reaction *vs.* temperature then you can determine the values for the enthalpy and entropy of activation for the single transition state in that single step.

They look the same, don't they. Both include a pre-exponential term and exponential terms. For a simple one-step reaction we could state that the Arrhenius plot and the Eyring plot must result in the same answer at a given temperature and the two plots must have the same slope (both plots must give the same line). Therefore, the derivative w.r.t. T (the x-axis) in both linear plots must be the same.

$$k = A \cdot e^{\frac{-E_a}{RT}} = \kappa \frac{k_B}{h} T \cdot e^{\frac{\Delta S^\ddagger}{R}} \cdot e^{\frac{-\Delta H^\ddagger}{RT}}$$

$$\ln k = \ln A - \frac{E_a}{RT} = \ln \left(\kappa \frac{k_B}{h} \right) + \ln T + \frac{\Delta S^\ddagger}{R} - \frac{\Delta H^\ddagger}{RT}$$

$$\frac{\partial}{\partial T} \ln A - \frac{\partial}{\partial T} \frac{E_a}{RT} = \frac{\partial}{\partial T} \ln \left(\kappa \frac{k_B}{h} \right) + \frac{\partial}{\partial T} \ln T + \frac{\partial}{\partial T} \frac{\Delta S^\ddagger}{R} - \frac{\partial}{\partial T} \frac{\Delta H^\ddagger}{RT}$$

$$0 + \frac{E_a}{RT^2} = 0 + \frac{1}{T} + 0 + \frac{\Delta H^\ddagger}{RT^2}$$

$$\frac{E_a}{RT^2} = \frac{1}{T} + \frac{\Delta H^\ddagger}{RT^2}$$

$$E_a = RT + \Delta H^\ddagger$$

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¹ See ch. 7.2

² The Arrhenius equation

$$k = A \cdot e^{\frac{-E_a}{RT}}$$

and the linear form

$$\ln k = \ln A - \frac{E_a}{RT}$$

³ The Eyring equation

$$k = \kappa \frac{k_B}{h} T \cdot e^{\frac{\Delta S^\ddagger}{R}} \cdot e^{\frac{-\Delta H^\ddagger}{RT}}$$

and the linear form

$$\ln k = \ln \left(\kappa \frac{k_B}{h} \right) + \ln T + \frac{\Delta S^\ddagger}{R} - \frac{\Delta H^\ddagger}{RT}$$

Equate the Arrhenius and the Eyring equations.

Some calculus review for you...

$$\frac{\partial}{\partial x} c = 0$$

$$\frac{\partial}{\partial x} cx = c \frac{\partial}{\partial x} x$$

$$\frac{\partial}{\partial x} \ln x = \frac{1}{x}$$

$$\frac{\partial}{\partial x} x^n = nx^{n-1}$$

$$\frac{\partial}{\partial x} \frac{1}{x} = -\frac{1}{x^2}$$

multiplied both sides by RT^2

We now have a term for E_a in single-step simple reactions that can be used to make the Arrhenius equation the same as the Eyring equation.

$$A \cdot e^{-\frac{(RT+\Delta H^\ddagger)}{RT}} = \kappa \frac{k_B}{h} T \cdot e^{\frac{\Delta S^\ddagger}{R}} \cdot e^{\frac{-\Delta H^\ddagger}{RT}}$$

$$A \cdot e^{-\frac{RT}{RT}} e^{\frac{-\Delta H^\ddagger}{RT}} = \kappa \frac{k_B}{h} T \cdot e^{\frac{\Delta S^\ddagger}{R}} \cdot e^{\frac{-\Delta H^\ddagger}{RT}}$$

$$A \cdot e^{-1} = \kappa \frac{k_B}{h} T \cdot e^{\frac{\Delta S^\ddagger}{R}}$$

$$A = \kappa \frac{k_B}{h} e \cdot T \cdot e^{\frac{\Delta S^\ddagger}{R}}$$

We now have a term for the Arrhenius pre-exponential value, A , in terms of $\Delta S^\ddagger$. Let us calculate the value of $\log A$ and see how this can lead us to that rule of thumb.

$$\log A = \log \kappa \frac{k_B}{h} e + \log T + \frac{\Delta S^\ddagger}{R} \cdot \log e$$

If we set $\Delta S^\ddagger = 0$ and work at room temperature, we see that...

$$\log A = \log \left(1 \cdot \frac{1.381 \times 10^{-23}}{6.626 \times 10^{-34}} \cdot 2.7183 \right) + \log 298 + \frac{0}{8.314 \times 2.303} = 13.2$$

At room temperature, we observe that values of $\log A$ that are less than 13.2 would have a negative entropy of activation and values that are more than 13.2 would have a positive entropy of activation. This dividing line is temperature-dependant but does not change much over usual laboratory temperatures (e.g., $\log 298 = 2.5$ and $\log 600 = 2.8$).

A rule of thumb is not a law. This chemical aphorism provides a quick way to evaluate the nature of the transition state but always remember that it may not give the correct answer.⁴ It is best used for simple first and second order reactions in solution. With those caveats in mind, here is the official rule of thumb:

For uncomplicated reactions in solution, we can estimate the $\Delta S^\ddagger$ using the following rule. If $\log A > 10.8 + \log T$ then the $\Delta S^\ddagger$ is positive; If $\log A < 10.8 + \log T$ then the $\Delta S^\ddagger$ is negative.

We can solve for $\Delta S^\ddagger$ if we wish and then see how different $\log A$ values reveal the value of the entropy of activation.

$$\Delta S^\ddagger = \frac{R}{\log e} \left(\log A - \log \kappa \frac{k_B}{h} e - \log T \right)$$

The value of $\Delta S^\ddagger$ increases in magnitude by $2.3R$ (approx. $20 \text{ J/mole}\cdot\text{K}$) for each log unit that we are away from the rule of thumb value for $\log A$ as demonstrate in Table ??.

Some math review...

$$e^{a+b} = e^a e^b$$

$e^{\frac{-\Delta H^\ddagger}{RT}}$ is on both sides and cancels

multiplied both sides by e

More math review...

$$\ln e^x = x$$

$$\log e^x = x \log e$$

$$\log e = \frac{1}{2.303}$$

Euler's number	$e = 2.7183$
Boltzmann const.	$k_B = 1.381 \times 10^{-23} \text{ J/K}$
Planck's const.	$h = 6.626 \times 10^{-34} \text{ J/s}$
Transmission coeff.	$\kappa = 1$
Gas const.	$R = 8.314 \text{ J/mole}\cdot\text{K}$

⁴ For more information on this topic see "The 'Yard Stick' to Interpret the Entropy of Activation in Chemical Kinetics: A Physical-Organic Chemistry Exercise", R. Sanjeev, D. A. Padmavathi, V. Jagannadham, *World Journal of Chemical Education*, 2018, 6, 78-81. <https://doi.org/10.12691/wjce-6-1-12>

Table 1: $\Delta S^\ddagger$ values at $T = 298\text{K}$

$\log A$	$\Delta S^\ddagger$
11.2	$-38 \text{ J/mole}\cdot\text{K}$
12.2	-19
13.2	near 0
14.2	19
15.2	38